

High-resolution mapping of gaseous heterogeneity and data-driven reduction of sampling density in a naturally ventilated dairy barn

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Abstract

Representative gas concentrations are required for reliable evaluation of air quality and gaseous emissions from naturally ventilated dairy barns, although the sampling density required to characterise their heterogeneous airspace remains uncertain. Concentrations of carbon dioxide (CO₂), methane (CH₄), and ammonia (NH₃) were measured at 51 internal locations arranged as 17 vertical triplets during a high-resolution campaign. Two cavity ring-down spectrometers and two Fourier-transform infrared spectrometers were operated at a temporal resolution of four minutes. Fourier-transform infrared measurements were harmonised to the cavity ring-down spectroscopy scale using correction factors obtained from a preliminary 24 h co-located comparison. Alternative vertical sampling configurations were evaluated in 178 complete two-hour blocks. Top-only sampling overestimated the full-density mean, whereas bottom-only sampling underestimated it. Combining the top and bottom heights reduced the network from 51 to 34 locations while preserving the mean concentrations of all three gases. Lin's concordance correlation coefficient between the full and reduced designs was 0.999 for CO₂, CH₄, and NH₃; the corresponding mean relative biases were below 0.4%. The reduced design was subsequently applied during a longer second campaign. The results demonstrate that sampling density can be reduced substantially when vertically contrasting locations are retained, whereas measurements at one height alone produce systematic gas-specific bias.

Keywords: air quality; dairy housing; measurement uncertainty; sensor placement; tracer ratio; vertical profile

Nomenclature

Symbol	Definition
CH ₄	methane
CO ₂	carbon dioxide
CCC	Lin's concordance correlation coefficient
CRDS	cavity ring-down spectroscopy
FTIR	Fourier-transform infrared spectroscopy
H ₂ O	water vapour
NH ₃	ammonia
NVDB	naturally ventilated dairy barn
RE	relative deviation from the full-density estimate, %
RMSE	root mean square error
<i>s</i>	south-background sampling location
SE	standard error
SP	sampling point

1 Introduction

Concentrations measured in naturally ventilated livestock buildings are affected by unsteady wind-driven exchange, local sources, animal activity, building geometry, and imperfect internal mixing. Consequently, a concentration recorded at one position cannot automatically be assumed to represent the complete barn airspace. This representativeness is important both for characterising animal exposure and for calculating gaseous emissions with indirect balance methods.

Spatial variability has been demonstrated in both the horizontal and vertical directions. In a naturally ventilated dairy cow barn, Mendes et al. (2015) reported height-dependent concentrations of NH₃, CO₂, and a released tracer gas, with mixing ratios becoming more stable above the animal-occupied zone. Measurements at three vertical locations in an open dairy barn also showed gas-specific profiles and only moderate correlations between some heights (D'Urso et al., 2024). Horizontal representativeness is similarly dependent on both the number and placement of sampling points. D'Urso et al. (2022) found that variability decreased when the number of horizontally distributed locations increased and that the distance between locations affected the resulting mean.

Increasing sampling density can reduce uncertainty, but it also increases equipment requirements, tubing, switching-cycle duration, data-processing effort, and the probability of sampling-system failure. In a naturally ventilated animal mock-up building, De Voogeleer et al. (2017) showed that the spatial pattern of sensors could be as influential as the number of sensors. Systematic reduction of a high-resolution wind-tunnel measure-

ment grid similarly showed that concentration and velocity sensors positioned at only one height could produce large emission errors, whereas vertically composite sampling improved representativeness (Janke et al., 2020). More recently, information-based sensor selection has been shown to retain most predictive performance with substantially fewer strategically selected locations (Chen et al., 2026).

Available studies therefore indicate that dense sampling is valuable, but do not establish how a full-scale, multi-height concentration network can be reduced without losing the barn mean and its spatial heterogeneity under field conditions. The objective of this study was to quantify the spatial heterogeneity of CO_2 , CH_4 , and NH_3 in a naturally ventilated dairy barn and to test the hypothesis that a spatially distributed top-and-bottom configuration would reproduce the concentration statistics obtained from a 51-location, three-height reference design.

2 Materials and methods

2.1 Experimental barn

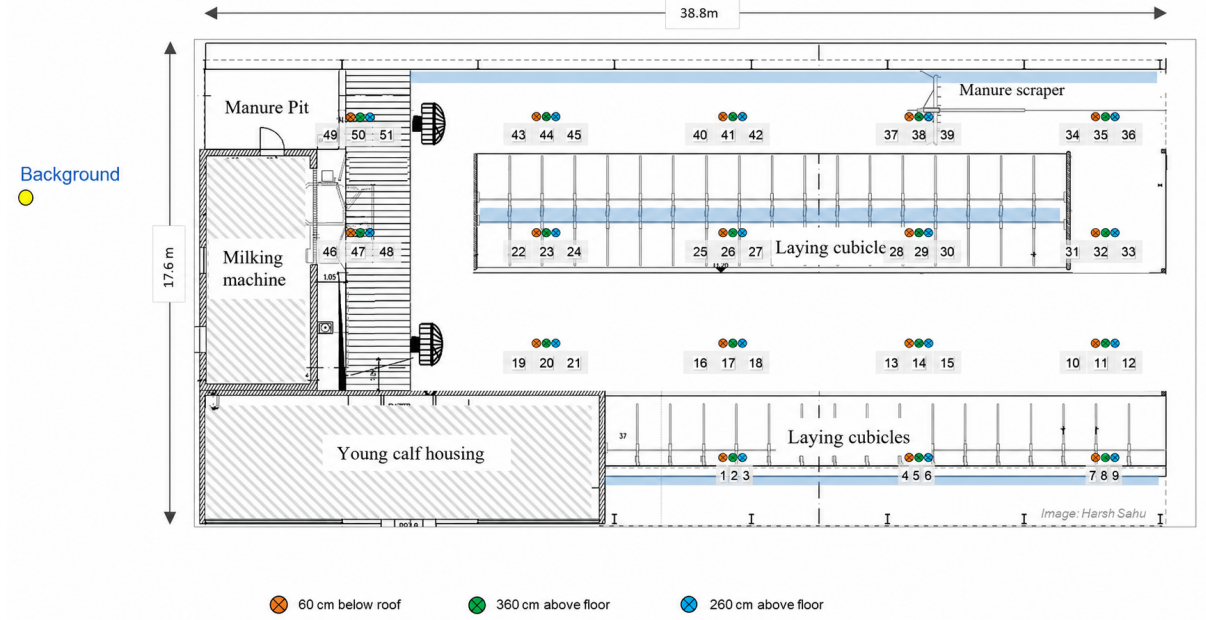
Measurements were conducted in the experimental dairy barn of the Lehr- und Versuchsanstalt für Tierzucht und Tierhaltung, Gross Kreutz, Brandenburg, Germany. The naturally ventilated barn was 38.88 m long and 17.65 m wide. The asymmetrical fibre-cement roof reached approximately 6.2 m at the ridge and approximately 3.6 m at the sidewalls. The internal airspace was approximately 4529 m^3 . The loose-housing barn was designed for 58 dairy cows and contained littered cubicles, concrete walkways, a manure scraper, and adjacent milking and young-stock areas.

2.2 Sampling layout

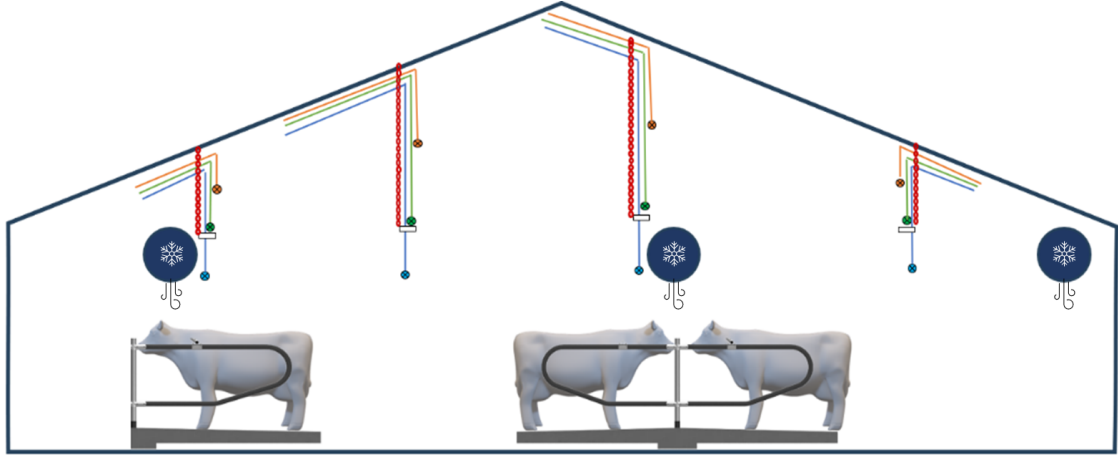
Fifty-one internal SPs were distributed over 17 horizontal positions (Fig. 1). Three vertically aligned locations were installed at each position: a top location 60 cm below the roof, a middle location 360 cm above the floor, and a bottom location 260 cm above the floor. The location numbers were assigned consecutively within each triplet. Thus, locations 1, 4, $\dots$, 49 represented the top group; locations 2, 5, $\dots$, 50 represented the middle group; and locations 3, 6, $\dots$, 51 represented the bottom group. A single south-background SP, denoted s , was treated conceptually as location 52 but was excluded from all analyses of internal spatial heterogeneity.

2.3 Gas analysers and preliminary field comparison

Concentrations of CO_2 , CH_4 , NH_3 , and water vapour were measured using two CRDS analysers (CRDS8 and CRDS9) and two FTIR analysers (FTIR1 and FTIR2). Before the



(a) Plan view.



(b) Cross-sectional representation of the sampling heights and fans.

Figure 1: Experimental barn and high-resolution sampling design. Orange symbols indicate locations 60 cm below the roof, green symbols indicate locations 360 cm above the floor, and blue symbols indicate locations 260 cm above the floor. The numbered internal locations comprise 17 vertical triplets. The yellow point labelled Background represents the south-background location s , which was retained in the dataset but excluded from internal spatial comparisons.

first mapping campaign, all four analysers were operated for 24 h at a co-located central barn position. Two polytetrafluoroethylene sampling tubes were used: one was divided with a T-piece between the two CRDS analysers and the other was divided between the two FTIR analysers. The FTIR instruments measured CO₂ and CH₄ concentrations that were 6% higher than those measured by CRDS, while FTIR NH₃ was 9% higher. Accordingly, FTIR measurements from Campaign 1 were harmonised to the CRDS scale as

$$C_{\text{CO}_2,\text{corr}} = C_{\text{CO}_2,\text{raw}}/1.06, \quad (1)$$

$$C_{\text{CH}_4,\text{corr}} = C_{\text{CH}_4,\text{raw}}/1.06, \quad (2)$$

$$C_{\text{NH}_3,\text{corr}} = C_{\text{NH}_3,\text{raw}}/1.09. \quad (3)$$

CRDS measurements and all water-vapour measurements were retained without correction. Both raw and corrected concentrations were preserved in the analytical dataset. Before and after the field comparison, all analysers were tested with certified cylinders containing approximately 700 ppm CO₂, 7 ppm CH₄, and 5 ppm NH₃; deviations remained within the manufacturer-stated $\pm 5\%$ accuracy. The field comparison was therefore used for cross-platform harmonisation under representative sampling conditions.

2.4 Measurement campaigns and data processing

Campaign 1 was conducted from 1 October 2024 at 00:02:32 to 24 October 2024 at 14:38:58. CRDS9 sampled locations 1–16, FTIR2 sampled locations 17–26, CRDS8 sampled locations 27–42, and FTIR1 sampled locations 43–51 and *s*. The resulting network comprised 51 internal locations and the south-background location. Gas measurements were recorded at a four-minute temporal resolution.

Campaign 2 was conducted from 16 November 2024 at 00:06:47 to 31 December 2024 at 23:57:10 using CRDS8 and CRDS9. The 17 middle locations were omitted following the Campaign 1 density assessment. Locations 19 and 40 were also excluded because of fan proximity and a sampling-line leakage problem, respectively, resulting in 32 internal locations. Each CRDS sampling step lasted 240 s; the first 60 s was discarded for flushing and the remaining 180 s was averaged.

Non-positive gas records were treated as missing in the manuscript analytical dataset. Carbon dioxide values below 300 ppm were also treated as invalid transition or analyser records because they were physically implausible for barn or ambient air. No row was deleted from the immutable source dataset.

Table 1: Measurement design and analytical records retained for the two campaigns. The south-background location s was retained in the data but excluded from the count of internal locations.

Campaign	Period	Analysers	Vertical design	Internal SPs	Rows
1	1–24 Oct. 2024	CRDS8, CRDS9, FTIR1, FTIR2	top, middle, bottom	51	31 997
2	16 Nov.–31 Dec. 2024	CRDS8, CRDS9	top and bottom	32	20 632

2.5 Gas ratios

The relative mixing ratios of CH_4 and NH_3 to CO_2 were calculated for positive, valid concentrations:

$$R_{\text{CH}_4:\text{CO}_2} = 100 C_{\text{CH}_4}/C_{\text{CO}_2}, \quad (4)$$

$$R_{\text{NH}_3:\text{CO}_2} = 100 C_{\text{NH}_3}/C_{\text{CO}_2}. \quad (5)$$

The ratios are dimensionless and are reported as percentages. Where inferential modelling of ratios is subsequently performed, their logarithmic forms, $\log(C_{\text{CH}_4}/C_{\text{CO}_2})$ and $\log(C_{\text{NH}_3}/C_{\text{CO}_2})$, are preferred to avoid the asymmetry of untransformed ratios.

2.6 Evaluation of sampling-density reduction

Daily means at each location were first calculated to provide independent blocks for descriptive means and 95% confidence intervals. For direct comparison of sampling configurations, measurements were averaged into two-hour blocks. A block was retained only when all 51 internal locations were represented, yielding 178 complete blocks.

Seven vertical configurations were reconstructed from the Campaign 1 data: top only, middle only, bottom only, top plus middle, middle plus bottom, top plus bottom, and the full three-height design. The full 51-location mean was used as the reference estimate within each block. For a reduced configuration k , the relative deviation was calculated as

$$RE_{k,t} = 100 \frac{\overline{C}_{k,t} - \overline{C}_{51,t}}{\overline{C}_{51,t}}. \quad (6)$$

The top-and-bottom configuration was selected a priori as the operational reduced design because it retained both ends of the observed vertical gradient. Its agreement with the full design was quantified using mean bias, relative bias, RMSE, coefficient of determination, and Lin’s CCC. Spatial heterogeneity within each block was expressed as the standard deviation of log-transformed location concentrations.

All cleaning, calculations, and figures were produced in R version 4.5.3 (R Foundation

for Statistical Computing, Vienna, Austria). Data manipulation was conducted using `data.table`, and figures were produced using `ggplot2`. Top, middle, and bottom locations were displayed in orange, green, and light blue, respectively.

3 Results

3.1 High-resolution concentration patterns

Marked spatial and vertical variation was detected during Campaign 1 (Fig. 2). Across internal locations, the mean corrected CO₂ concentrations were approximately 670 ppm at the top, 652 ppm at the middle, and 639 ppm at the bottom. Corresponding CH₄ means were 21.3, 20.2, and 19.2 ppm, while NH₃ means were 1.83, 1.72, and 1.61 ppm. The direction and magnitude of the vertical gradient were not uniform among horizontal positions. Particularly distinct values occurred at locations 19 and 40, supporting their exclusion from the subsequent operational design.

3.2 Effect of vertical sampling configuration

Use of one height alone produced systematic deviations from the full-density mean (Fig. 3; Table 2). Top-only sampling overestimated CO₂, CH₄, and NH₃ by mean values of 2.7%, 6.0%, and 8.4%, respectively. Bottom-only sampling underestimated the same gases by 2.3%, 5.3%, and 7.8%. The middle-only configuration was closer to the full mean, but removed the vertical extremes required for evaluating heterogeneity.

Combining top and bottom locations substantially reduced configuration bias. The median absolute deviations were 0.33% for CO₂, 0.79% for CH₄, and 0.81% for NH₃. Their 95th percentiles remained below 2.6% for all gases.

3.3 Agreement between full and reduced designs

The top-and-bottom mean closely reproduced the full 51-location estimate throughout Campaign 1 (Fig. 4). Lin's CCC was 0.999 for all three gases. Mean relative biases were 0.19% for CO₂, 0.34% for CH₄, and 0.27% for NH₃. The corresponding RMSE values were 3.02 ppm, 0.24 ppm, and 0.018 ppm. Agreement for the spatial heterogeneity metric was lower than agreement for the mean but remained strong, with CCC values between 0.953 and 0.968.

3.4 Longer-term application of the reduced design

The reduced-height principle was applied from mid-November to the end of December 2024 in Campaign 2 (Fig. 5). Mean concentrations remained higher at top than at bot-

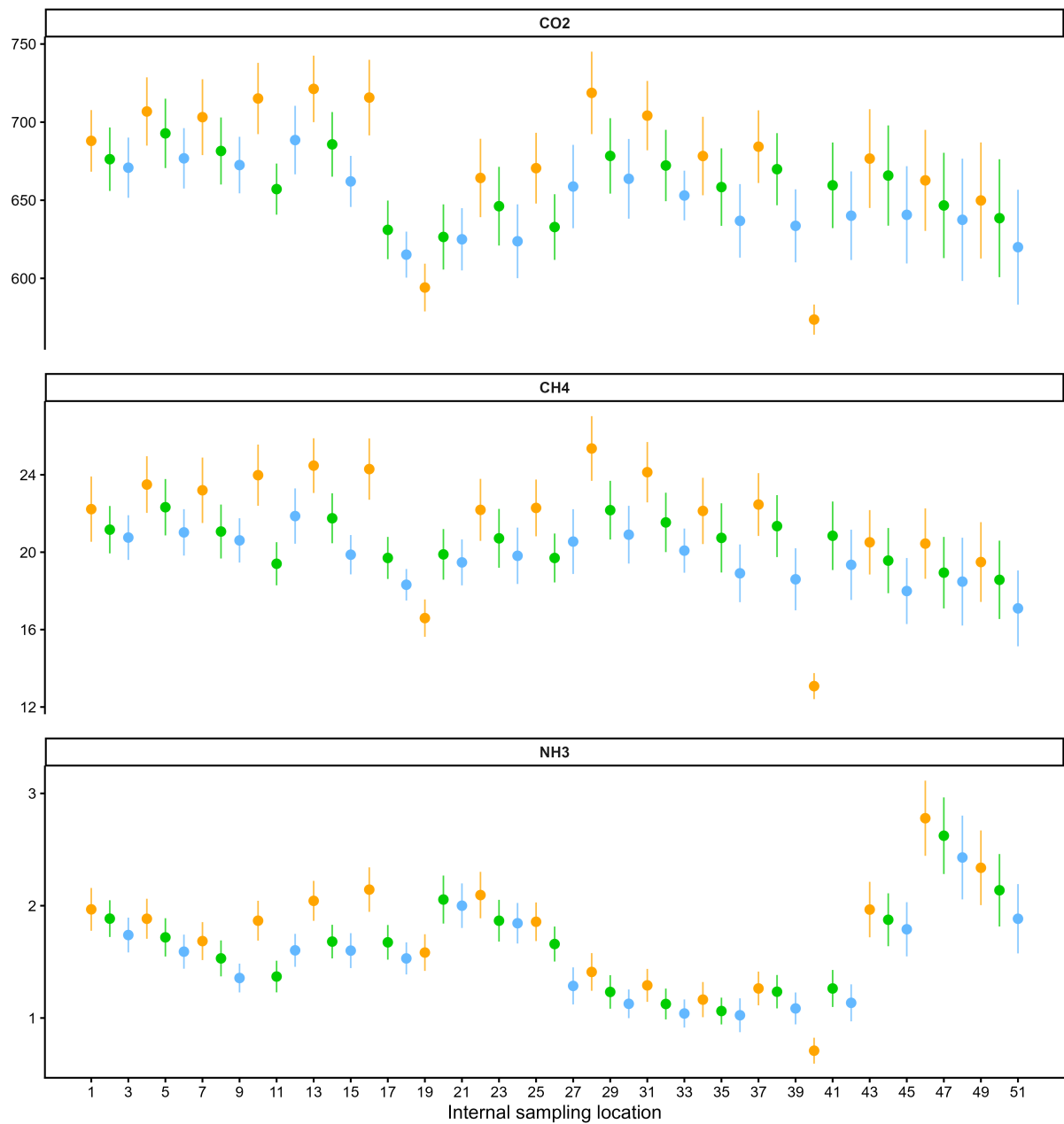


Figure 2: Campaign 1 corrected concentrations at the 51 internal sampling locations. Points denote means of daily location means and vertical bars denote 95% confidence intervals calculated across daily blocks. Orange, green, and light-blue symbols denote top, middle, and bottom locations, respectively. The south-background location s was excluded.

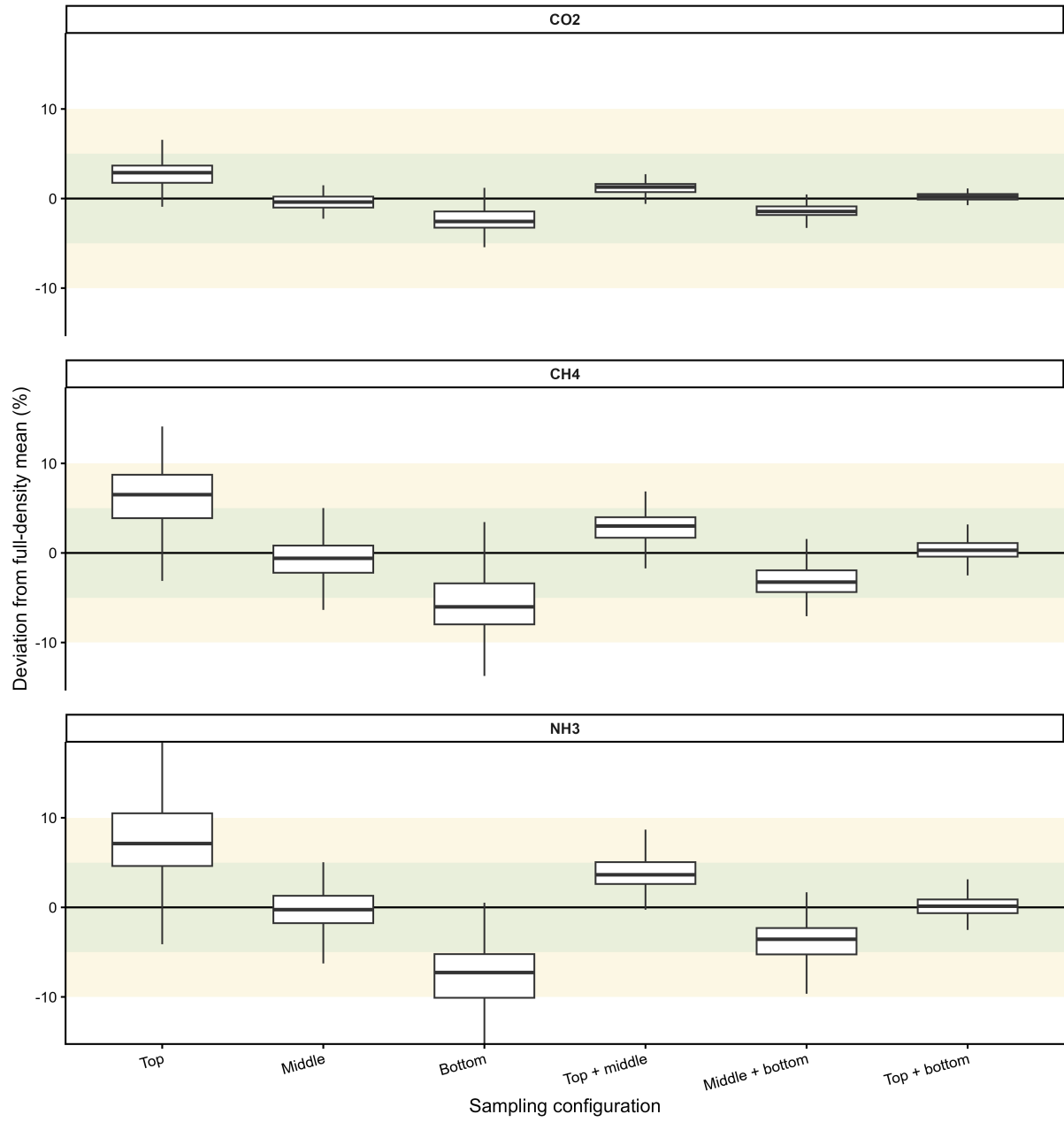


Figure 3: Deviation of six reduced vertical sampling configurations from the full 51-location mean in 178 complete two-hour blocks. Boxes show the interquartile range and centre lines show medians; whiskers extend to 1.5 times the interquartile range. Green and yellow bands indicate deviations within $\pm 5\%$ and $\pm 10\%$, respectively. Configurations labelled top, middle, or bottom contain 17 locations; paired-height configurations contain 34 locations. Outliers are omitted visually but retained in all summaries.

Table 2: Performance of selected reduced configurations relative to the full 51-location mean. Values are calculated across 178 complete two-hour blocks. MAD denotes median absolute deviation and P_{95} denotes the 95th percentile of absolute deviation.

Gas	Configuration	Mean RE (%)	MAD (%)	P_{95} (%)
CO ₂	Top only	2.68	2.89	4.91
	Middle only	-0.38	0.65	1.83
	Bottom only	-2.30	2.57	4.19
	Top + bottom	0.19	0.33	0.92
CH ₄	Top only	6.01	6.65	12.12
	Middle only	-0.68	1.58	5.02
	Bottom only	-5.33	6.15	10.55
	Top + bottom	0.34	0.79	2.51
NH ₃	Top only	8.35	7.13	18.19
	Middle only	-0.54	1.63	4.97
	Bottom only	-7.81	7.29	15.80
	Top + bottom	0.27	0.81	2.48

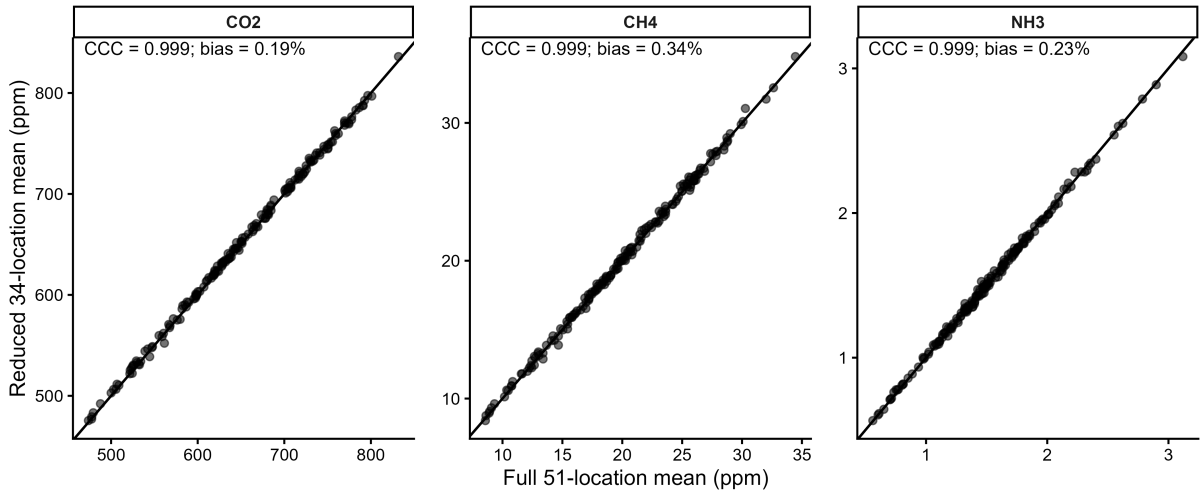


Figure 4: Agreement between the full 51-location mean and the reduced 34-location top-and-bottom mean in 178 complete two-hour blocks. Solid lines represent exact 1:1 agreement. CCC denotes Lin's concordance correlation coefficient, and bias is the mean relative difference from the full-density estimate.

tom locations: CO₂ averaged approximately 640 and 625 ppm, CH₄ averaged 18.6 and 17.4 ppm, and NH₃ averaged 0.96 and 0.83 ppm at the top and bottom, respectively. The mean CH₄/CO₂ ratios were 2.75% and 2.64%, and the mean NH₃/CO₂ ratios were 0.149% and 0.131% at the top and bottom.

4 Discussion

4.1 Concentration heterogeneity and vertical structure

The full-scale measurements confirmed that the assumption of homogeneous gas concentrations was not supported throughout the barn volume. Concentrations were generally greater at the top than at the bottom, but the gradient varied among horizontal positions and gases. This gas-specific behaviour agrees with the vertical variability reported by Mendes et al. (2015) and D’Urso et al. (2024). Those studies showed that vertical profiles cannot be generalised from one gas to another and that correlations among heights may remain incomplete. The present network extended this evidence from a limited number of vertical locations to 17 horizontally distributed triplets.

The differences among locations also support the horizontal-distribution findings of D’Urso et al. (2022), who showed that an individual location was unlikely to provide a representative barn mean and that both the number and separation of locations affected variability. In the present barn, the contrast among the 17 horizontal positions was sufficiently large that vertical reduction was only successful when spatial coverage across the barn was retained.

4.2 Why the top-and-bottom configuration was retained

The middle-only configuration gave a comparatively small mean bias, but it did not preserve the vertical extremes. Selection of a network solely because it reproduces the overall mean would therefore conceal part of the spatial heterogeneity that the measurement system was intended to characterise. In contrast, the combined top-and-bottom configuration balanced the positive deviation at the top against the negative deviation at the bottom while retaining sensitivity to vertical gradients.

This result is consistent with the general finding that sensor placement is not interchangeable with sensor number. De Voeleer et al. (2017) demonstrated that configurations containing the same number of sensors could yield different accuracy because of their spatial pattern. Janke et al. (2020) reported particularly large errors when concentration and velocity were represented by one height and improved performance from vertically composite sampling. The present field measurements provide an analogous result for internal barn concentrations: retaining two vertically contrasting levels was more representative than retaining either level alone.

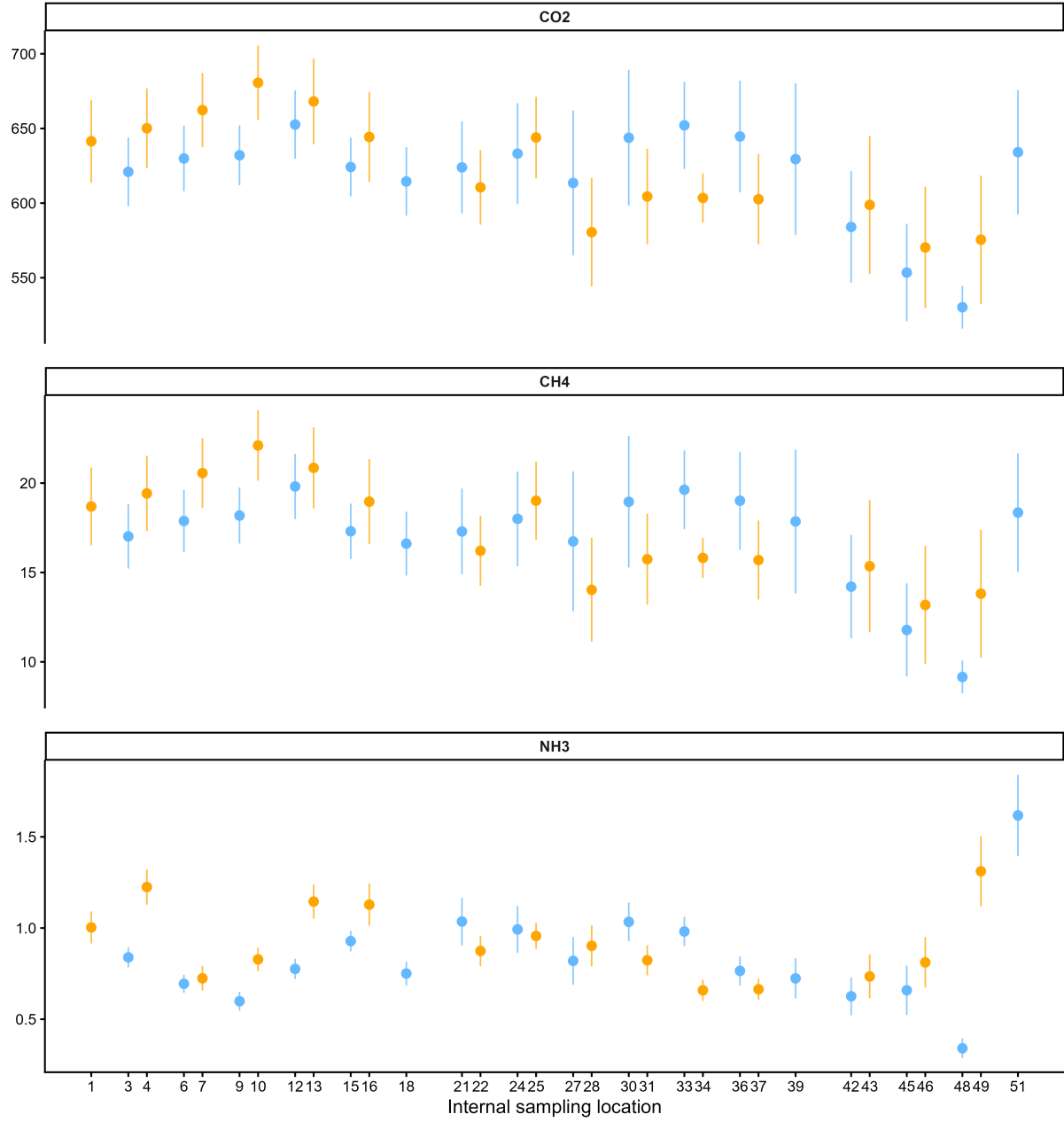


Figure 5: Campaign 2 concentrations measured with the reduced top-and-bottom network. Points denote means of daily location means and vertical bars denote 95% confidence intervals calculated across daily blocks. Orange and light-blue symbols denote top and bottom locations, respectively.

The near-unity CCC for the 34-location mean demonstrates that the reduction introduced little loss of information about the overall concentration. Nevertheless, agreement for the spatial heterogeneity metric was weaker than agreement for the mean. This distinction is important because a reduced network may reproduce the mean through compensating positive and negative deviations while slightly changing its description of spatial spread. Accordingly, both outcomes should be evaluated when sampling networks are optimised.

4.3 Implications for sensor-network optimisation

The reduction from 51 to 34 locations lowered the number of internal sampling points by one third without materially changing the two-hour barn mean. Campaign 2 subsequently showed that the reduced-height principle remained operational over a longer measurement period. This progression follows the sensor-network concept advanced by Chen et al. (2026): a full network can first be used to identify informative spatial structure, after which a smaller network can be retained for routine monitoring.

The design should not, however, be transferred directly to barns with different geometry or ventilation without validation. Optimal positions are expected to depend on openings, internal obstructions, source distribution, fans, wind direction, and the intended response variable. The present findings support a general procedure—dense initial mapping followed by objective reduction—rather than a universal requirement for exactly 34 locations.

4.4 Limitations and future research

Several limitations should be considered. First, Campaign 1 was conducted in October and Campaign 2 in late autumn and winter; consequently, the complete range of seasonal ventilation conditions was not represented. Second, the four analysers were assigned to different location ranges. Although the preliminary co-located comparison was used to harmonise FTIR and CRDS values, analyser type and barn location were not completely separable during Campaign 1. Third, sequential switching meant that all locations were not sampled at the exact same second. Two-hour complete blocks were therefore used to compare configurations. Fourth, the 51-location estimate was treated as the full-density reference rather than as an absolute true barn concentration.

The Campaign 2 network contained 32 rather than 34 locations because two locations were removed for identified site-specific problems. Therefore, Campaign 2 provided a longer-term application of the reduced principle but not an independent 51-versus-34 validation. Future investigations should combine simultaneous meteorological, animal, and barn-climate measurements with independent validation of strategically reduced networks.

5 Conclusions

Gas concentrations in the investigated naturally ventilated dairy barn were heterogeneous in both the vertical and horizontal directions. Measurements from only the top systematically overrepresented the full-density mean, whereas measurements from only the bottom systematically underrepresented it. The middle height reproduced the overall mean more closely but did not retain the observed vertical extremes.

A spatially distributed top-and-bottom network preserved the mean CO₂, CH₄, and NH₃ concentrations obtained from the full 51-location design while reducing the number of internal locations by one third. Dense initial mapping followed by quantitative configuration testing provides a defensible route for reducing sampling effort without assuming homogeneous mixing.

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Author contributions

Harsh Sahu: conceptualisation, methodology, investigation, data curation, formal analysis, visualisation, and writing—original draft. Long Chen: methodology, formal analysis, and writing—review and editing. Thomas Amon: conceptualisation, supervision, funding acquisition, and writing—review and editing. Tariq Mehmood: investigation and writing—review and editing. Aditya Rawat: investigation and writing—review and editing. David Janke: conceptualisation, methodology, supervision, project administration, and writing—review and editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The processed data and analysis code will be made available in a public repository upon acceptance. *[Repository and DOI to be added.]*

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